Lesson 11: Static and Extensive-Form Games

Nash Equilibrium, Minimax Theorem, and Subgame Perfection

Session 11 of 102 | Phase I: Mathematical Foundations | 8 h
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Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

Abstract

This lesson develops the foundational theory of strategic interaction: games in which multiple decision-makers choose actions simultaneously or sequentially, each seeking to maximize their own payoff. We define normal-form games, prove Nash's existence theorem for mixed-strategy equilibria via Kakutani's fixed point theorem, establish von Neumann's minimax theorem for zero-sum games, introduce correlated equilibrium, and then pass to extensive-form games where we define subgame perfect equilibrium and prove Zermelo's theorem by backward induction. Every theorem is proved in full. The reader who masters this material will possess the game-theoretic language and solution concepts that underpin all multi-agent reinforcement learning algorithms in Phase 05.

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1 Why Game Theory for RL and ML?

In single-agent reinforcement learning, the environment is typically modeled as a Markov decision process (MDP): the agent chooses actions, the environment responds according to a fixed (if unknown) transition kernel, and the agent seeks a policy maximizing cumulative reward. The key simplification is that the environment does not *adapt* to the agent's behavior.

Multi-agent reinforcement learning (MARL) breaks this assumption. When two or more learning agents interact in a shared environment, each agent's "environment" includes the other agents, whose behavior changes as they learn. The resulting non-stationarity means that standard single-agent convergence guarantees fail. To analyze MARL, we need a mathematical framework for strategic interaction — and that framework is game theory.

MARL Is Game Theory + Learning

Every MARL algorithm implicitly assumes a game-theoretic solution concept:

- **Independent Q-learning**: each agent performs Q-learning ignoring others. If it converges, the result is a Nash equilibrium of the stage game (under restrictive conditions).
- **Minimax-Q** (Littman, 1994): assumes a two-player zero-sum stochastic game and seeks the minimax value — directly applying von Neumann's theorem (Theorem 6.2).
- **Nash-Q** (Hu & Wellman, 2003): computes Nash equilibria of stage games at each step.
- **MAPPO, QMIX**: cooperative MARL algorithms whose convergence analysis relies on the team game structure (common payoff games).

Without understanding Nash equilibria, minimax, and subgame perfection, one cannot state what these algorithms are trying to compute, let alone prove convergence.

This lesson covers three progressively richer models of strategic interaction:

1. **Normal-form (static) games** (Sections 2–7): players choose actions simultaneously.
2. **Extensive-form (dynamic) games** (Sections 8–9): players move sequentially, possibly with imperfect information.
3. **Solution concepts**: Nash equilibrium, minimax strategies, correlated equilibrium, and subgame perfect equilibrium.

Prerequisites. Familiarity with basic probability (finite probability spaces suffice for this lesson), linear algebra (convex sets, hyperplane separation), and real analysis (continuity, compactness in $\mathbb{R}^n$). Knowledge of linear programming duality is helpful but not required; we state the needed result.

2 Normal-Form Games

2.1 Basic Definitions

Definition 2.1 (Normal-Form Game)**.** A *normal-form game* (or *strategic-form game*) is a tuple $\Gamma = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ where:

(i) $N = \{1, 2, \ldots, n\}$ is a finite set of *players*.

(ii) For each $i \in N$, S_i is a non-empty set of *strategies* (or *actions*) available to player i. We write $S = \prod_{i\in N} S_i$ for the set of *strategy profiles*.

(iii) For each $i \in N$, $u_i : S \to \mathbb{R}$ is player i's *payoff function* (or *utility function*).

The game is *finite* if each S_i is a finite set.

Remark 2.1 (Notation for "All but i"). For a strategy profile $s = (s_1, \ldots, s_n) \in S$, we write $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n) \in S_{-i} := \prod_{j \neq i} S_j$. We often write $s = (s_i, s_{-i})$ to emphasize player i's strategy.

Definition 2.2 (Bimatrix Game)**.** A *bimatrix game* is a two-player finite game presented as a pair of matrices $(A, B) \in \mathbb{R}^{m \times n} \times \mathbb{R}^{m \times n}$, where player 1 has m strategies, player 2 has n strategies, $A_{jk} = u_1(j, k)$, and $B_{jk} = u_2(j, k)$. If $B = -A$, the game is *zero-sum*.

2.2 Classical Examples

Example 2.1 (Prisoner's Dilemma)**.** Two players simultaneously choose to *Cooperate* (C) or *Defect* (D):

	C	D
C	$(3, 3)$	$(0, 5)$
D	$(5, 0)$	$(1, 1)$

Here D strictly dominates C for both players (Definition 2.3), so the unique Nash equilibrium is (D, D) with payoffs $(1, 1)$, despite (C, C) giving both players a higher payoff.

Example 2.2 (Battle of the Sexes)**.** Players 1 and 2 prefer different activities (Opera vs. Football) but prefer coordination to miscoordination:

	O	F
O	$(3, 1)$	$(0, 0)$
F	$(0, 0)$	$(1, 3)$

This game has two pure Nash equilibria — (O, O) and (F, F) — and one mixed equilibrium (computed in Example 4.1).

Example 2.3 (Matching Pennies)**.** A zero-sum game: each player chooses Heads or Tails. Player 1 wins if they match; player 2 wins if they differ:

	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

This game has no pure Nash equilibrium (Proposition 3.2), motivating the extension to mixed strategies.

Example 2.4 (Stag Hunt)**.** Players choose to hunt a Stag (requiring cooperation) or a $H$are (safe but less rewarding):

	S	H
S	$(4, 4)$	$(0, 3)$
H	$(3, 0)$	$(3, 3)$

Both (S, S) and (H, H) are pure Nash equilibria. The equilibrium (S, S) is *payoff-dominant* (higher payoffs for both), while (H, H) is *risk-dominant* (safer against uncertainty about the opponent).

Stag Hunt and Cooperative MARL

The stag hunt captures a fundamental tension in cooperative MARL: agents can achieve higher collective reward through coordination, but each agent bears risk if the others fail to coordinate. This is precisely the challenge faced by CTDE (centralized training, decentralized execution) algorithms like QMIX and MAPPO.

2.3 Dominant and Dominated Strategies

Definition 2.3 (Dominance)**.** Let $\Gamma = (N, (S_i), (u_i))$ be a normal-form game and fix player i. A strategy $s_i \in S_i$ *strictly dominates* $s_i' \in S_i$ if

$$u_i(s_i, s_{-i}) > u_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

A strategy s_i *weakly dominates* s_i' if

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i},$$

with strict inequality for at least one s_{-i}. A strategy is *strictly dominated* if some other strategy strictly dominates it. A strategy is a *(strictly) dominant strategy* if it strictly dominates every other strategy of that player.

Proposition 2.1 (Dominated Strategies Are Never Best Responses)**.** *If s_i' is strictly dominated by s_i, then s_i' is never a best response: for every $s_{-i} \in S_{-i}$, $s_i' \notin \operatorname{argmax}_{t_i \in S_i} u_i(t_i, s_{-i})$.*

Proof. Fix any $s_{-i} \in S_{-i}$. Since s_i strictly dominates s_i', we have $u_i(s_i, s_{-i}) > u_i(s_i', s_{-i})$, so s_i' does not maximize $u_i(\cdot, s_{-i})$. □

2.4 Iterated Elimination of Strictly Dominated Strategies

Definition 2.4 (IESDS)**.** *Iterated elimination of strictly dominated strategies* (IESDS) is the following procedure. Set $S_i^0 = S_i$ for all i. For $k = 0, 1, 2, \ldots$, define

$$S_i^{k+1} = \left\{ s_i \in S_i^k : \text{there is no } s_i' \in S_i^k \text{ such that } u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}) \text{ for all } s_{-i} \in S_{-i}^k \right\}.$$

The set of *surviving strategies* for player i is $S_i^\infty = \bigcap_{k=0}^\infty S_i^k$.

Theorem 2.2 (Order Independence of IESDS)**.** *In a finite game, the set of strategy profiles surviving iterated elimination of strictly dominated strategies is independent of the order of elimination.*

Proof. We prove that any strategy eliminated in some order of elimination is eventually eliminated in every order.

Let s_i^* be a strategy of player i that is eliminated at round k in some particular elimination order E. This means there exists $s_i' \in S_i^{k-1}(E)$ such that $u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$ for all $s_{-i} \in S_{-i}^{k-1}(E)$.

Consider an alternative elimination order E'. We show s_i^* is eventually eliminated under E'. Let $T = S_{-i}^\infty(E')$ be the surviving opponents' strategies under E'.

Since $S_{-i}^\infty(E') \subseteq S_{-i}^{k-1}(E')$ and every strategy surviving under any order is in particular not eliminated in the first $k-1$ rounds under E (we prove this by induction on rounds), we need a cleaner argument.

We proceed by induction on the round of elimination. For the base case ($k = 0$), s_i^* is strictly dominated in the original game, so it will be eliminated in round 0 under any order.

For the inductive step, suppose every strategy eliminated in rounds $0, \ldots, k-1$ under *any* elimination order is eventually eliminated under every order. Now suppose s_i^* is eliminated at round k under order E. Then there exists $s_i' \in S_i^{k-1}(E)$ with $u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$ for all $s_{-i} \in S_{-i}^{k-1}(E)$.

Under order E', the set $S_{-i}^{k-1}(E)$ may differ from $S_{-i}^{k-1}(E')$. However, the key observation is:

Claim: $S_{-i}^\infty(E') \subseteq S_{-i}^{k-1}(E)$.

To prove the claim, note that any strategy s_j removed before round k under order E was strictly dominated by some s_j' against strategies surviving at that point. By the inductive

hypothesis, s_j is eventually removed under E' as well, so $s_j \notin S_j^\infty(E')$. Taking contrapositives: $s_j \in S_j^\infty(E') \implies s_j \in S_j^{k-1}(E)$, establishing the claim.

Now, since $S_{-i}^\infty(E') \subseteq S_{-i}^{k-1}(E)$, the domination $u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$ holds for all $s_{-i} \in S_{-i}^\infty(E')$. If $s_i' \in S_i^\infty(E')$, then s_i^* is dominated (by s_i') relative to the surviving strategies under E' and hence is eliminated. If s_i' is itself eliminated under E', then s_i' was dominated by some s_i'', and by transitivity of strict domination on the surviving set, s_i^* remains dominated and is eliminated.

By induction, the surviving sets coincide for all elimination orders. □

2.5 Rationalizability

Definition 2.5 (Best Response)**.** Player i's *best response correspondence* is the set-valued map $\mathrm{BR}_i : S_{-i} \rightrightarrows S_i$ defined by

$$\mathrm{BR}_i(s_{-i}) = \operatorname{argmax}_{s_i \in S_i} u_i(s_i, s_{-i}).$$

Definition 2.6 (Rationalizable Strategies)**.** A strategy s_i is *rationalizable* if it survives the iterated elimination of strategies that are never best responses. Formally, set $R_i^0 = S_i$ and define

$$R_i^{k+1} = \left\{ s_i \in R_i^k : \exists\, s_{-i} \in R_{-i}^k \text{ such that } s_i \in \mathrm{BR}_i(s_{-i}) \right\}.$$

The set of rationalizable strategies is $R_i^\infty = \bigcap_{k=0}^\infty R_i^k$.

Proposition 2.3 (IESDS and Rationalizability in Two-Player Games)**.** *In finite two-player games, the set of rationalizable strategies coincides with the set of strategies surviving IESDS.*

Proof. Every strictly dominated strategy is never a best response (Proposition 2.1), so $S_i^\infty \supseteq R_i^\infty$. Conversely, in a two-player finite game, a strategy that is never a best response to any pure strategy of the opponent is strictly dominated (this follows from the supporting hyperplane theorem applied to the payoff polyhedron). Hence a strategy eliminated in the rationalizability procedure is strictly dominated relative to the surviving set, giving $R_i^\infty \supseteq S_i^\infty$. □

Three or More Players

For games with three or more players, IESDS may not eliminate all never-best-response strategies. Rationalizability can be a strictly coarser concept than IESDS survival in games with $n \geq 3$ players when we restrict to pure strategies. The equivalence is restored when we extend dominance to allow domination by mixed strategies.

3 Nash Equilibrium in Pure Strategies

Definition 3.1 (Pure Strategy Nash Equilibrium)**.** A strategy profile $s^* = (s_1^*, \ldots, s_n^*) \in S$ is a *(pure strategy) Nash equilibrium* (NE) if no player can unilaterally improve their payoff:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for all } s_i \in S_i, \text{ for all } i \in N.$$

Equivalently, $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for all $i \in N$.

Proposition 3.1 (NE as Mutual Best Response)**.** *A strategy profile s^* is a Nash equilibrium if and only if $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for every $i \in N$.*

Proof. By Definition 3.1, s^* is a NE if and only if $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$ for all $s_i \in S_i$ and all i. By Definition 2.5, this is precisely $s_i^* \in \operatorname{argmax}_{s_i \in S_i} u_i(s_i, s_{-i}^*) = \mathrm{BR}_i(s_{-i}^*)$ for all i. □

Proposition 3.2 (Matching Pennies Has No Pure NE)**.** *The Matching Pennies game (Example 2.3) has no pure strategy Nash equilibrium.*

Proof. We check all four profiles:

- (H, H): Player 2 gains by deviating to T (payoff $-1 \to 1$).
- (H, T): Player 1 gains by deviating to T (payoff $-1 \to 1$).
- (T, H): Player 1 gains by deviating to H (payoff $-1 \to 1$).
- (T, T): Player 2 gains by deviating to H (payoff $-1 \to 1$).

Every profile admits a profitable deviation, so no pure NE exists. □

Remark 3.1. The Prisoner's Dilemma has a unique NE at (D, D), the Battle of the Sexes has two pure NE at (O, O) and (F, F), and the Stag Hunt has two at (S, S) and (H, H). Matching Pennies has none — motivating the extension to mixed strategies.

4 Mixed Strategies and Mixed Extensions

4.1 Definitions

Definition 4.1 (Mixed Strategy)**.** Let Γ be a finite game with strategy set $S_i = \{s_i^1, \ldots, s_i^{m_i}\}$ for player i. A *mixed strategy* for player i is a probability distribution $\sigma_i \in \Delta(S_i)$, where

$$\Delta(S_i) = \left\{ \sigma_i \in \mathbb{R}^{m_i} : \sigma_i(s_i^j) \geq 0 \text{ for all } j, \ \sum_{j=1}^{m_i} \sigma_i(s_i^j) = 1 \right\}$$

is the $(m_i - 1)$-*simplex* over S_i. A *mixed strategy profile* is $\sigma = (\sigma_1, \ldots, \sigma_n) \in \Delta := \prod_{i \in N} \Delta(S_i)$.

Definition 4.2 (Support)**.** The *support* of a mixed strategy σ_i is $\mathrm{supp}(\sigma_i) = \{s_i \in S_i : \sigma_i(s_i) > 0\}$. A pure strategy s_i is identified with the degenerate distribution placing probability 1 on s_i.

Definition 4.3 (Mixed Extension)**.** The *mixed extension* of a finite game $\Gamma = (N, (S_i), (u_i))$ is the game $\bar{\Gamma} = (N, (\Delta(S_i)), (\bar{u}_i))$ where the expected payoff under independent mixing is

$$\bar{u}_i(\sigma) = \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j) \right) u_i(s) = \mathbb{E}_{s \sim \sigma}[u_i(s)].$$

We drop the bar and write $u_i(\sigma)$ when the context is clear.

Remark 4.1 (Multilinearity). The expected payoff $u_i(\sigma)$ is *multilinear*: it is linear in each σ_i when the other players' strategies are held fixed. Explicitly,

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i) \underbrace{\sum_{s_{-i} \in S_{-i}} \left(\prod_{j \neq i} \sigma_j(s_j) \right) u_i(s_i, s_{-i})}_{=: u_i(s_i, \sigma_{-i})}.$$

The quantity $u_i(s_i, \sigma_{-i})$ is the expected payoff to player i from playing the pure strategy s_i when opponents play σ_{-i}.

4.2 Mixed Nash Equilibrium and the Indifference Principle

Definition 4.4 (Mixed Strategy Nash Equilibrium)**.** A mixed strategy profile σ^* is a *Nash equilibrium* if

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*) \quad \text{for all } \sigma_i \in \Delta(S_i), \text{ for all } i \in N.$$

Theorem 4.1 (Indifference Principle)**.** *Let σ^* be a mixed strategy Nash equilibrium. For each player i, every pure strategy in* $\text{supp}(\sigma_i^*)$ *yields the same expected payoff against σ_{-i}^*, and this common payoff is at least as large as the expected payoff from any pure strategy outside the support. Formally:*

(a) If $s_i, s_i' \in \text{supp}(\sigma_i^*)$, then $u_i(s_i, \sigma_{-i}^*) = u_i(s_i', \sigma_{-i}^*)$.

(b) If $s_i \in \text{supp}(\sigma_i^*)$ and $t_i \notin \text{supp}(\sigma_i^*)$, then $u_i(s_i, \sigma_{-i}^*) \geq u_i(t_i, \sigma_{-i}^*)$.

Proof. By multilinearity (Remark 4.1),

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \sum_{s_i \in S_i} \sigma_i^*(s_i)\, u_i(s_i, \sigma_{-i}^*).$$

Let $v_i^* = u_i(\sigma_i^*, \sigma_{-i}^*)$ be the equilibrium payoff for player i.

Part (b): Suppose $t_i \in S_i$ satisfies $u_i(t_i, \sigma_{-i}^*) > v_i^*$. Then the pure strategy t_i (viewed as a degenerate mixed strategy) gives payoff $u_i(t_i, \sigma_{-i}^*) > v_i^* = u_i(\sigma_i^*, \sigma_{-i}^*)$, contradicting the NE condition. Hence $u_i(s_i, \sigma_{-i}^*) \leq v_i^*$ for all s_i with equality impossible for all s_i unless we also have a lower bound. Since v_i^* is a convex combination of the payoffs $u_i(s_i, \sigma_{-i}^*)$ with weights $\sigma_i^*(s_i) \geq 0$ summing to 1, if any payoff in the support were strictly less than v_i^*, then some payoff (in or outside the support) must be strictly greater than v_i^* (to make the weighted average equal v_i^*), which we just showed is impossible. Therefore:

- For all $s_i \in \text{supp}(\sigma_i^*)$: $u_i(s_i, \sigma_{-i}^*) = v_i^*$.
- For all $s_i \notin \text{supp}(\sigma_i^*)$: $u_i(s_i, \sigma_{-i}^*) \leq v_i^*$.

Part (a): Immediate from the first bullet point. □

Example 4.1 (Mixed NE in Battle of the Sexes)**.** In the Battle of the Sexes (Example 2.2), let player 1 play O with probability p and player 2 play O with probability q. By the indifference principle, in a fully mixed NE, player 1 must be indifferent:

$$u_1(O, q) = u_1(F, q) \implies 3q + 0(1-q) = 0 \cdot q + 1 \cdot (1-q) \implies 3q = 1 - q \implies q = \tfrac{1}{4}.$$

Similarly, player 2 must be indifferent:

$$u_2(p, O) = u_2(p, F) \implies 1 \cdot p + 0(1-p) = 0 \cdot p + 3(1-p) \implies p = 3 - 3p \implies p = \tfrac{3}{4}.$$

The mixed NE is $\sigma_1 = (\frac{3}{4}, \frac{1}{4})$, $\sigma_2 = (\frac{1}{4}, \frac{3}{4})$, with expected payoffs $(3 \cdot \frac{1}{4}, 1 \cdot \frac{3}{4}) = (\frac{3}{4}, \frac{3}{4})$. Both players fare worse in the mixed NE than in either pure NE.

Mixed Strategies and Stochastic Policies

In RL, a stochastic policy $\pi(a|s)$ assigns probabilities to actions given a state. This is precisely a mixed strategy in the stage game at state s. The indifference principle has a direct analog: in an optimal stochastic policy, every action in the support must yield the same Q-value. If some action had a strictly higher Q-value, the agent should place all probability on it. This is why ϵ-greedy exploration is suboptimal — it assigns probability to actions that may not be in the support of any Nash equilibrium.

5 Nash's Existence Theorem

The central result of this lesson: every finite game has at least one Nash equilibrium in mixed strategies. The proof uses Kakutani's fixed point theorem.

5.1 Kakutani's Fixed Point Theorem

Definition 5.1 (Upper Hemicontinuity)**.** Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$. A correspondence (set-valued map) $F : X \rightrightarrows Y$ is *upper hemicontinuous* (uhc) at $x_0 \in X$ if for every open set $V \supseteq F(x_0)$, there exists an open set $U \ni x_0$ such that $F(x) \subseteq V$ for all $x \in U \cap X$.

Equivalently (when Y is compact), F is uhc at x_0 if whenever $x_k \to x_0$ and $y_k \in F(x_k)$ with $y_k \to y_0$, we have $y_0 \in F(x_0)$. This is the *closed graph* characterization.

Theorem 5.1 (Kakutani's Fixed Point Theorem)**.** *Let $K \subseteq \mathbb{R}^n$ be non-empty, compact, and convex. Let $F : K \rightrightarrows K$ be a correspondence such that:*

(i) $F(x)$ is non-empty for every $x \in K$.
(ii) $F(x)$ is convex for every $x \in K$.
(iii) F is upper hemicontinuous.

Then F has a fixed point: there exists $x^* \in K$ with $x^* \in F(x^*)$.

Proof. We provide the standard proof via Brouwer's fixed point theorem.

For each $k \in \mathbb{N}$, let T_k be a triangulation of K with mesh size at most $1/k$ (the maximum diameter of any simplex in the triangulation). For each vertex v of T_k, choose $f_k(v) \in F(v)$ (possible since $F(v)$ is non-empty). Extend f_k to all of K by convex interpolation on each simplex: if $x = \sum_j \lambda_j v_j$ is the barycentric representation of x in its simplex, set $f_k(x) = \sum_j \lambda_j f_k(v_j)$.

We claim each $f_k : K \to K$ is continuous. Continuity within each simplex follows from the linearity of the interpolation. On shared faces of adjacent simplices, the interpolation agrees because barycentric coordinates are unique on the shared face. Since K is convex and each $f_k(x)$ is a convex combination of points in K, we have $f_k(x) \in K$.

By Brouwer's fixed point theorem (applicable since K is compact and convex, and f_k is continuous), each f_k has a fixed point $x_k \in K$ with $f_k(x_k) = x_k$.

Since K is compact, the sequence (x_k) has a convergent subsequence $x_{k_l} \to x^* \in K$.

We show $x^* \in F(x^*)$. For each l, let σ_l be the simplex of T_{k_l} containing x_{k_l}, with vertices $v_1^l, \ldots, v_p^l$ (where $p \leq n+1$). Then

$$x_{k_l} = f_{k_l}(x_{k_l}) = \sum_{j=1}^{p} \lambda_j^l f_{k_l}(v_j^l), \quad \text{where } x_{k_l} = \sum_{j=1}^{p} \lambda_j^l v_j^l.$$

Since the mesh size tends to 0, $\|v_j^l - x_{k_l}\| \leq 1/k_l \to 0$, so $v_j^l \to x^*$ for each j.

Now, $f_{k_l}(v_j^l) \in F(v_j^l)$. Since K is compact, by passing to a further subsequence (using a diagonal argument over the finitely many vertices), we may assume $f_{k_l}(v_j^l) \to y_j$ for some $y_j \in K$.

By the closed-graph characterization of upper hemicontinuity (Definition 5.1): $v_j^l \to x^*$ and $f_{k_l}(v_j^l) \to y_j$ with $f_{k_l}(v_j^l) \in F(v_j^l)$ implies $y_j \in F(x^*)$.

The barycentric coordinates $\lambda_j^l \in [0,1]$ lie in a compact set, so (passing to a further subsequence) $\lambda_j^l \to \lambda_j^*$ with $\sum_j \lambda_j^* = 1$ and $\lambda_j^* \geq 0$.

Taking the limit in $x_{k_l} = \sum_j \lambda_j^l f_{k_l}(v_j^l)$:

$$x^* = \sum_j \lambda_j^* y_j.$$

Since each $y_j \in F(x^*)$ and $F(x^*)$ is convex, the convex combination $x^* = \sum_j \lambda_j^* y_j \in F(x^*)$. $\square$

5.2 Nash's Theorem

Nash's Existence Theorem

Theorem 5.2 (Nash, 1950). *Every finite normal-form game has at least one mixed strategy Nash equilibrium.*

Proof. Let $\Gamma = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ be a finite game with $|S_i| = m_i$. Define the set of mixed strategy profiles $\Delta = \prod_{i\in N} \Delta(S_i)$, which is a product of simplices and hence compact and convex in $\mathbb{R}^{m_1+\cdots+m_n}$.

Step 1: Define the best response correspondence. For each player i, the *mixed best response correspondence* $\mathrm{BR}_i : \Delta_{-i} \rightrightarrows \Delta(S_i)$ is

$$\mathrm{BR}_i(\sigma_{-i}) = \mathrm{argmax}_{\sigma_i\in\Delta(S_i)}\, u_i(\sigma_i, \sigma_{-i}).$$

Define the joint best response correspondence $\mathrm{BR} : \Delta \rightrightarrows \Delta$ by

$$\mathrm{BR}(\sigma) = \mathrm{BR}_1(\sigma_{-1}) \times \mathrm{BR}_2(\sigma_{-2}) \times \cdots \times \mathrm{BR}_n(\sigma_{-n}).$$

Step 2: $\mathrm{BR}(\sigma)$ is non-empty. For each i, the function $\sigma_i \mapsto u_i(\sigma_i, \sigma_{-i})$ is continuous (in fact, linear) on the compact set $\Delta(S_i)$. By the extreme value theorem, the maximum is attained, so $\mathrm{BR}_i(\sigma_{-i}) \neq \emptyset$.

Step 3: $\mathrm{BR}(\sigma)$ is convex. Fix i and σ_{-i}. Let $\sigma_i, \sigma_i' \in \mathrm{BR}_i(\sigma_{-i})$ and $\lambda \in [0, 1]$. Set $\hat{\sigma}_i = \lambda\sigma_i + (1 - \lambda)\sigma_i'$. By linearity of u_i in σ_i (Remark 4.1):

$$u_i(\hat{\sigma}_i, \sigma_{-i}) = \lambda\, u_i(\sigma_i, \sigma_{-i}) + (1 - \lambda)\, u_i(\sigma_i', \sigma_{-i}) = \lambda\, v_i^* + (1 - \lambda)\, v_i^* = v_i^*,$$

where $v_i^* = \max_{\tau_i} u_i(\tau_i, \sigma_{-i})$. So $\hat{\sigma}_i$ also achieves the maximum, hence $\hat{\sigma}_i \in \mathrm{BR}_i(\sigma_{-i})$. Thus $\mathrm{BR}_i(\sigma_{-i})$ is convex. Since $\mathrm{BR}(\sigma)$ is a product of convex sets, it is convex.

Step 4: BR is upper hemicontinuous. We use the closed-graph characterization. Let $\sigma^k \to \sigma^*$ in Δ and $\tau^k \in \mathrm{BR}(\sigma^k)$ with $\tau^k \to \tau^*$. We must show $\tau^* \in \mathrm{BR}(\sigma^*)$, i.e., $\tau_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$ for each i.

Fix player i. Since $\tau_i^k \in \mathrm{BR}_i(\sigma_{-i}^k)$, for every $\sigma_i \in \Delta(S_i)$:

$$u_i(\tau_i^k, \sigma_{-i}^k) \geq u_i(\sigma_i, \sigma_{-i}^k).$$

Since u_i is continuous (it is multilinear on the compact set Δ), taking $k \to \infty$:

$$u_i(\tau_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*).$$

Since this holds for all σ_i, we have $\tau_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$.

Step 5: Apply Kakutani. The correspondence $\mathrm{BR} : \Delta \rightrightarrows \Delta$ satisfies all hypotheses of Kakutani's theorem (Theorem 5.1): Δ is non-empty, compact, and convex; $\mathrm{BR}(\sigma)$ is non-empty and convex for each σ; and BR is upper hemicontinuous. Therefore, there exists $\sigma^* \in \Delta$ with $\sigma^* \in \mathrm{BR}(\sigma^*)$.

By definition of BR, $\sigma_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$ for all i, which by Proposition 3.1 (extended to mixed strategies) means σ^* is a Nash equilibrium. □

Remark 5.1 (Alternative Proof via Brouwer). An alternative proof avoids Kakutani by constructing a continuous function whose fixed points are NE. For each player i and pure strategy s_i^j, define the "excess payoff":

$$g_i^j(\sigma) = \max\left\{0,\ u_i(s_i^j, \sigma_{-i}) - u_i(\sigma_i, \sigma_{-i})\right\}.$$

Then define $f : \Delta \to \Delta$ by

$$f_i^j(\sigma) = \frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)}.$$

One verifies: (i) f maps Δ to Δ; (ii) f is continuous; (iii) σ^* is a fixed point of f if and only if $g_i^j(\sigma^*) = 0$ for all i, j, which holds if and only if σ^* is a NE. Brouwer's theorem gives a fixed point, hence a NE.

Existence Guarantees in MARL

Nash's theorem guarantees that every finite MARL problem (modeled as a stochastic game with finite state and action spaces) has at least one Nash equilibrium — at least in the stage games. This justifies the goal of Nash-Q learning (Hu & Wellman, 2003), which attempts to compute Nash equilibria at each state. However, the theorem is non-constructive: it guarantees existence but says nothing about efficient computation. Indeed, computing a Nash equilibrium is PPAD-complete (Daskalakis, Goldberg, Papadimitriou, 2009), meaning no polynomial-time algorithm is known.

6 Von Neumann's Minimax Theorem

6.1 Two-Player Zero-Sum Games

Definition 6.1 (Two-Player Zero-Sum Game)**.** A two-player game $(N, (S_1, S_2), (u_1, u_2))$ is *zero-sum* if $u_1(s) + u_2(s) = 0$ for all $s \in S$. It is specified by a single payoff matrix $A \in \mathbb{R}^{m \times n}$ where $u_1(i, j) = A_{ij}$ and $u_2(i, j) = -A_{ij}$. Player 1 (the "row player") maximizes; player 2 (the "column player") minimizes.

Definition 6.2 (Maximin and Minimax Values)**.** For a zero-sum game with payoff matrix A, define:

$$\underline{v} = \max_{x \in \Delta_m} \min_{y \in \Delta_n} x^\top A y \quad \text{(maximin value: player 1's guarantee)},$$
$$\overline{v} = \min_{y \in \Delta_n} \max_{x \in \Delta_m} x^\top A y \quad \text{(minimax value: player 2's guarantee)},$$

where Δ_k denotes the $(k-1)$-simplex in $\mathbb{R}^k$.

Lemma 6.1 (Maximin $\leq$ Minimax)**.** *For any matrix $A \in \mathbb{R}^{m \times n}$, $\underline{v} \leq \overline{v}$.*

Proof. For any fixed $x \in \Delta_m$ and $y \in \Delta_n$:

$$\min_{y' \in \Delta_n} x^\top A y' \leq x^\top A y \leq \max_{x' \in \Delta_m} (x')^\top A y.$$

Taking max over x on the left and min over y on the right:

$$\underline{v} = \max_x \min_{y'} x^\top A y' \leq \min_y \max_{x'} (x')^\top A y = \overline{v}.$$ □

Von Neumann's Minimax Theorem

Theorem 6.2 (Von Neumann, 1928)**.** *For every matrix $A \in \mathbb{R}^{m \times n}$,*

$$\max_{x \in \Delta_m} \min_{y \in \Delta_n} x^\top A y = \min_{y \in \Delta_n} \max_{x \in \Delta_m} x^\top A y.$$

The common value $v^* = \underline{v} = \overline{v}$ is called the value *of the game.*

Proof. We give a proof using LP duality. By Lemma 6.1, it suffices to show $\underline{v} \geq \overline{v}$.

Step 1: Reformulate as linear programs. Player 1's maximin problem is:

$$\underline{v} = \max_{x \in \Delta_m} \min_{y \in \Delta_n} x^\top A y.$$

Since $y \mapsto x^\top A y$ is linear in y and Δ_n is a simplex, the minimum over y is attained at a vertex e_j (the j-th standard basis vector), giving $\min_y x^\top A y = \min_{j=1,\ldots,n} (A^\top x)_j$.

Thus $\underline{v} = \max_{x \in \Delta_m} \min_j (A^\top x)_j$. Introducing an auxiliary variable v, this is equivalent to the linear program:

$$\begin{aligned} \text{(P)} \quad & \max \ v \\ & \text{s.t.} \ A^\top x \geq v\,\mathbf{1}, \quad x \in \Delta_m, \quad v \in \mathbb{R}, \end{aligned} \tag{1}$$

where $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^n$ and the inequality is componentwise.

Step 2: The dual LP. Similarly, player 2's minimax problem gives:

$$\overline{v} = \min_{y \in \Delta_n} \max_i (Ay)_i.$$

This is equivalent to:

$$\begin{aligned} \text{(D)} \quad & \min \ w \\ & \text{s.t.} \ Ay \leq w\,\mathbf{1}, \quad y \in \Delta_n, \quad w \in \mathbb{R}. \end{aligned} \tag{2}$$

Step 3: LP duality. We show that (P) and (D) are LP duals of each other. Rewrite (P) in standard form. Let $v = \underline{v}$. The constraints of (P) are: $A^\top x - v\,\mathbf{1} \geq 0$, $\mathbf{1}^\top x = 1$, $x \geq 0$. This is a linear program in the variables $(x, v) \in \mathbb{R}^{m+1}$.

The dual of this LP (after careful bookkeeping of the dual variables associated with each primal constraint) yields precisely the LP (D). By strong LP duality (both problems are feasible — take $x = (1/m, \ldots, 1/m)^\top$ and $y = (1/n, \ldots, 1/n)^\top$ — and bounded), the optimal values are equal:

$$\underline{v} = \text{opt(P)} = \text{opt(D)} = \overline{v}. \qquad \square$$

Corollary 6.3 (NE in Zero-Sum Games)**.** *In a two-player zero-sum game with payoff matrix A and value v^*:*

(a) (x^*, y^*) is a Nash equilibrium if and only if x^* is a maximin strategy and y^* is a minimax strategy.

(b) All NE yield the same payoff: $u_1 = v^*$, $u_2 = -v^*$.

(c) NE strategies are interchangeable: if (x^*, y^*) and (x^{**}, y^{**}) are both NE, then so are (x^*, y^{**}) and (x^{**}, y^*).

Proof. **(a)** ($\Rightarrow$): Let (x^*, y^*) be a NE. Then x^* is a best response to y^*, so $(x^*)^\top A y^* = \max_x x^\top A y^*$. Similarly, y^* minimizes $x^\top A y$ against x^*, so $(x^*)^\top A y^* = \min_y (x^*)^\top A y$. Therefore:

$$\max_x x^\top A y^* = (x^*)^\top A y^* = \min_y (x^*)^\top A y.$$

But $\min_y (x^*)^\top A y \leq \underline{v} \leq \overline{v} \leq \max_x x^\top A y^*$, and by the minimax theorem $\underline{v} = \overline{v} = v^*$, so all inequalities are equalities. Hence $\min_y (x^*)^\top A y = v^*$ (so x^* achieves the maximin) and $\max_x x^\top A y^* = v^*$ (so y^* achieves the minimax).

($\Leftarrow$): If x^* achieves the maximin and y^* achieves the minimax, then

$$(x^*)^\top A y^* \geq \min_y (x^*)^\top A y = v^* = \max_x x^\top A y^* \geq (x^*)^\top A y^*,$$

so $(x^*)^\top A y^* = v^*$. For any x: $x^\top A y^* \leq \max_{x'} (x')^\top A y^* = v^* = (x^*)^\top A y^*$, so x^* is a best response to y^*. Similarly, y^* is a best response to x^*. Hence (x^*, y^*) is a NE.

(b) Follows from part (a): every NE strategy pair achieves the value v^*.

(c) If x^* and x^{**} are both maximin and y^*, y^{**} both minimax, then by part (a), any combination of a maximin and a minimax strategy forms a NE. $\square$

Minimax in RL

The minimax theorem is the theoretical foundation of:

- **Minimax-Q learning** (Littman, 1994): at each state of a two-player zero-sum stochastic game, the Bellman operator computes $V(s) = \max_x \min_y x^\top Q(s, \cdot, \cdot)y$, which is well-defined by the minimax theorem.
- **AlphaGo/AlphaZero**: the MCTS search implicitly approximates minimax values in the game tree.
- **GANs**: the generator-discriminator game is a continuous zero-sum game $\min_G \max_D V(D, G)$. The minimax theorem (extended to continuous strategy spaces) justifies seeking saddle points.

6.2 Connection to Linear Programming

Proposition 6.4 (Solving Zero-Sum Games via LP). *The maximin strategy x^* and the game value v^* can be computed by solving the LP (1). The minimax strategy y^* is recovered from the dual LP (2). Both LPs have polynomial-time algorithms (e.g., interior point methods), so computing NE in two-player zero-sum games is in P.*

Proof. This follows directly from the reformulation in the proof of Theorem 6.2. The LP (1) has $m+1$ variables and $n+m+1$ constraints (including non-negativity and the simplex constraint), so it can be solved efficiently. □

Zero-Sum vs. General-Sum Complexity

While NE in two-player zero-sum games can be found in polynomial time via LP, computing a Nash equilibrium in a general two-player game is PPAD-complete (Chen, Deng, Teng, 2009; Daskalakis, Goldberg, Papadimitriou, 2009). This complexity-theoretic barrier is a fundamental challenge for MARL: even *finding* the solution concept that algorithms like Nash-Q target is computationally intractable in general.

7 Correlated Equilibrium

Definition 7.1 (Correlated Equilibrium). A *correlated equilibrium* (CE) of a finite game $\Gamma = (N, (S_i), (u_i))$ is a probability distribution π over $S = \prod_{i \in N} S_i$ such that for every player i and every pair of strategies $s_i, s_i' \in S_i$:

$$\sum_{s_{-i} \in S_{-i}} \pi(s_i, s_{-i})\, u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} \pi(s_i, s_{-i})\, u_i(s_i', s_{-i}).$$

Equivalently, if a mediator draws $s = (s_1, \ldots, s_n)$ from π and privately recommends s_i to player i, no player has an incentive to deviate from the recommendation (assuming the others follow theirs).

Theorem 7.1 (Every Nash Equilibrium Is a Correlated Equilibrium). *If $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed strategy NE, then the product distribution $\pi(s) = \prod_{i \in N} \sigma_i^*(s_i)$ is a correlated equilibrium.*

Proof. Fix player i and strategies $s_i, s_i' \in S_i$. The CE incentive constraint requires:

$$\sum_{s_{-i}} \pi(s_i, s_{-i})\, u_i(s_i, s_{-i}) \geq \sum_{s_{-i}} \pi(s_i, s_{-i})\, u_i(s_i', s_{-i}).$$

Since π is a product distribution, $\pi(s_i, s_{-i}) = \sigma_i^*(s_i) \cdot \prod_{j \neq i} \sigma_j^*(s_j)$. If $\sigma_i^*(s_i) = 0$, both sides are 0 and the inequality holds. If $\sigma_i^*(s_i) > 0$, dividing both sides by $\sigma_i^*(s_i)$ gives:

$$\sum_{s_{-i}} \left(\prod_{j \neq i} \sigma_j^*(s_j) \right) u_i(s_i, s_{-i}) \geq \sum_{s_{-i}} \left(\prod_{j \neq i} \sigma_j^*(s_j) \right) u_i(s_i', s_{-i}),$$

i.e., $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$. Since $s_i \in \operatorname{supp}(\sigma_i^*)$, the indifference principle (Theorem 4.1) gives $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$ for all s_i', which is exactly what we need. □

Proposition 7.2 (CE as a Linear Program)**.** *The set of correlated equilibria of a finite game is a convex polytope, defined by the linear incentive-compatibility constraints in Definition 7.1 together with $\pi(s) \geq 0$ for all s and $\sum_s \pi(s) = 1$. In particular, optimizing any linear objective over the CE set is a linear program.*

Proof. The CE constraints are:

$$\sum_{s_{-i}} \pi(s_i, s_{-i}) \big[u_i(s_i, s_{-i}) - u_i(s_i', s_{-i}) \big] \geq 0 \quad \text{for all } i, s_i, s_i'.$$

These are linear in the variables $\{\pi(s)\}_{s \in S}$. Combined with the simplex constraints $\pi(s) \geq 0$ and $\sum_s \pi(s) = 1$, the feasible set is a polytope (intersection of finitely many half-spaces). Any linear function of π can be optimized over this polytope by solving an LP. □

Example 7.1 (Correlated Equilibrium in Battle of the Sexes)**.** In the Battle of the Sexes (Example 2.2), consider the "traffic light" correlated equilibrium: a mediator recommends (O, O) with probability 1/2 and (F, F) with probability 1/2. Under this distribution π:

$$\pi(O, O) = 1/2, \quad \pi(O, F) = 0, \quad \pi(F, O) = 0, \quad \pi(F, F) = 1/2.$$

Check the CE condition for player 1, given recommendation O (so $s_1 = O$):

$$\pi(O, O)\, u_1(O, O) = \tfrac{1}{2} \cdot 3 = \tfrac{3}{2} \geq \pi(O, O)\, u_1(F, O) = \tfrac{1}{2} \cdot 0 = 0. \quad \checkmark$$

Given recommendation F:

$$\pi(F, F)\, u_1(F, F) = \tfrac{1}{2} \cdot 1 = \tfrac{1}{2} \geq \pi(F, F)\, u_1(O, F) = \tfrac{1}{2} \cdot 0 = 0. \quad \checkmark$$

Similarly for player 2. The expected payoffs are $(1/2)(3) + (1/2)(1) = 2$ for player 1 and $(1/2)(1) + (1/2)(3) = 2$ for player 2, which Pareto-dominates the mixed NE payoffs of $(3/4, 3/4)$.

Correlated Equilibrium and MARL

Correlated equilibrium has several advantages over Nash equilibrium for MARL:

- **Computational tractability**: finding a CE is an LP, polynomial in the game size. Finding a NE is PPAD-complete.
- **Natural emergence**: simple no-regret learning dynamics (e.g., regret matching) converge to the set of CE, not NE. This makes CE a more natural prediction for what independent learners will achieve.
- **Better welfare**: CE can achieve higher social welfare than any NE, as the Battle of the Sexes example illustrates.

8 Extensive-Form Games

8.1 Game Trees

Definition 8.1 (Extensive-Form Game)**.** An *extensive-form game* is a tuple $\Gamma = (N, H, P, (\mathcal{I}_i)_{i \in N}, u)$ where:

(i) $N = \{1, \ldots, n\}$ is the set of players. We may include a "chance" player 0 (nature).

(ii) H is a set of *histories* (finite sequences of actions), including the empty sequence $\emptyset$ (the root). H is closed under prefixes: if $(a_1, \ldots, a_k) \in H$ then $(a_1, \ldots, a_l) \in H$ for all $l \leq k$. A history $h \in H$ is *terminal* if there is no a with $(h, a) \in H$. The set of terminal histories is $Z \subseteq H$.

(iii) $P : H \setminus Z \to N \cup \{0\}$ is the *player function*, assigning to each non-terminal history the player who moves at that point.

(iv) For each player $i \in N$, $\mathcal{I}_i$ is a partition of $\{h \in H \setminus Z : P(h) = i\}$ into *information sets*. If h, h' are in the same information set $I \in \mathcal{I}_i$, then the set of available actions $A(h) = A(h')$ is the same (denoted $A(I)$).

(v) $u = (u_1, \ldots, u_n) : Z \to \mathbb{R}^n$ assigns payoffs to terminal histories.

Definition 8.2 (Perfect Information)**.** An extensive-form game has *perfect information* if every information set is a singleton: $|I| = 1$ for all $I \in \mathcal{I}_i$, for all $i \in N$. In a game with perfect information, every player knows exactly which node the game has reached when it is their turn to move.

Definition 8.3 (Perfect Recall)**.** Player i has *perfect recall* if, for any two histories h, h' in the same information set $I \in \mathcal{I}_i$, player i's sequence of information sets and actions leading to h and h' is the same. Informally, no player forgets their own past actions or information.

Example 8.1 (Entry Game)**.** An incumbent (player 2) faces a potential entrant (player 1). The entrant first decides whether to *Enter* (E) or *Stay Out* (O). If the entrant enters, the incumbent decides whether to *Fight* (F) or *Accommodate* (A):

- O: payoffs $(0, 2)$.
- (E, F): payoffs $(-1, -1)$.
- (E, A): payoffs $(1, 1)$.

This is a game with perfect information (all information sets are singletons).

8.2 Strategies in Extensive-Form Games

Definition 8.4 (Pure Strategy in Extensive Form)**.** A *pure strategy* for player i in an extensive-form game is a function $s_i : \mathcal{I}_i \to \bigcup_{I \in \mathcal{I}_i} A(I)$ satisfying $s_i(I) \in A(I)$ for each $I \in \mathcal{I}_i$.

Remark 8.1 (Strategies as Complete Contingent Plans). A strategy specifies an action at *every* information set, including those that cannot be reached given earlier choices. In the Entry Game (Example 8.1), player 2's strategy must specify what to do if the entrant enters, even if that node is never reached. This redundancy is essential for the normal-form representation and for defining subgame perfection.

Definition 8.5 (Behavioral Strategy)**.** A *behavioral strategy* for player i is a function $b_i : \mathcal{I}_i \to \bigcup_{I \in \mathcal{I}_i} \Delta(A(I))$ satisfying $b_i(I) \in \Delta(A(I))$. That is, b_i assigns a probability distribution over actions at each information set, with independent randomization at different information sets.

Proposition 8.1 (Normal-Form Representation)**.** *Every extensive-form game has a corresponding normal-form game: the pure strategy sets are as in Definition 8.4, and the payoff $u_i(s_1, \ldots, s_n)$ is determined by tracing the unique path through the game tree when each player follows their strategy (resolving chance moves by expectation).*

Proof. Given a strategy profile $(s_1, \ldots, s_n)$, the unique path from the root to a terminal node is determined inductively: at each non-terminal history h, the next action is $s_{P(h)}(I)$ where I is the information set containing h (or a probability distribution if $P(h) = 0$, i.e., nature moves). Since the game tree is finite, this process terminates at some terminal history $z \in Z$. Define $u_i(s_1, \ldots, s_n)$ as the expected value of $u_i(z)$ where the expectation is over any chance moves. The resulting tuple $(N, (S_i), (u_i))$ is a well-defined normal-form game. □

Theorem 8.2 (Kuhn's Theorem)**.** *In a finite extensive-form game with perfect recall, every mixed strategy of player i has a* realization-equivalent *behavioral strategy: a behavioral strategy that induces the same probability distribution over terminal histories for every strategy profile of the other players. Conversely, every behavioral strategy has a realization-equivalent mixed strategy.*

Proof. We prove the direction from mixed to behavioral, which is the more useful one.

Let σ_i be a mixed strategy (a distribution over pure strategies) for player i in a game with perfect recall. We construct a behavioral strategy b_i as follows.

For each information set $I \in \mathcal{I}_i$ and action $a \in A(I)$, define

$$b_i(a|I) = \frac{\sigma_i\big(\{s_i : s_i(I) = a \text{ and } s_i \text{ reaches } I\}\big)}{\sigma_i\big(\{s_i : s_i \text{ reaches } I\}\big)},$$

where "s_i reaches I" means that the pure strategy s_i, together with some strategy profile of the opponents, leads to a history in I. In a game with perfect recall, whether s_i reaches I depends only on s_i's earlier actions, so this is well-defined. If the denominator is zero (no pure strategy in the support of σ_i reaches I), set $b_i(\cdot|I)$ arbitrarily.

The key property of perfect recall is that it makes the randomization at different information sets conditionally independent: knowing that player i reaches information set I does not reveal what σ_i prescribes at other information sets that are not on the path to I. This ensures that the behavioral strategy b_i produces the same realization probabilities (probability of reaching each terminal node) as σ_i.

The converse (behavioral to mixed) follows by defining $\sigma_i(s_i) = \prod_I b_i(s_i(I)|I)$, viewing each pure strategy as specifying an action at each information set. □

Behavioral Strategies and RL Policies

A behavioral strategy is exactly an RL policy: it maps information states (information sets) to probability distributions over actions. Kuhn's theorem justifies the standard RL assumption that we can restrict attention to behavioral strategies (policies) without loss of generality. The perfect recall condition is the Markov property in disguise: the agent's policy can depend on the current information state without needing to recall the randomization device used to reach that state.

9 Subgame Perfect Equilibrium

9.1 Motivation: Non-Credible Threats

Example 9.1 (Non-Credible Threat in the Entry Game)**.** In the Entry Game (Example 8.1), consider the strategy profile where player 1 plays O (stay out) and player 2 plays F (fight). The payoffs are $(0, 2)$. Is this a NE?

Player 1's payoff from deviating to E (enter): given player 2's strategy F, the payoff is $-1 < 0$. So player 1 has no incentive to deviate.

Player 2's strategy F is not tested because player 1 stays out, so player 2's information set is not reached. Player 2 trivially has no incentive to deviate (any strategy gives the same outcome of $(0, 2)$ when player 1 plays O).

Therefore (O, F) is a NE. But it relies on a *non-credible threat*: player 2 threatens to fight, but if the entrant actually entered, fighting gives player 2 a payoff of -1 versus 1 for accommodating. Rational player 2 would never actually fight. The Nash equilibrium concept fails to exclude such non-credible threats because it does not require rational behavior at unreached information sets.

9.2 Subgames and Subgame Perfection

Definition 9.1 (Subgame)**.** A *subgame* of an extensive-form game Γ beginning at history $h \in H \setminus Z$ is the restriction of Γ to the histories that extend h, provided:

(i) h is a singleton information set: $\{h\}$ is an information set for player $P(h)$.

(ii) Every information set that contains a successor of h is entirely contained within the subtree rooted at h.

The subgame includes all histories $h' \in H$ such that h is a prefix of h', with the same player function, information sets (restricted to the subtree), and payoffs.

Remark 9.1. In a game with perfect information, every non-terminal history h begins a subgame (since all information sets are singletons). In games with imperfect information, subgames may be rare — the entire game is always a subgame (beginning at the root).

Definition 9.2 (Subgame Perfect Equilibrium)**.** A strategy profile s^* is a *subgame perfect equilibrium* (SPE) if its restriction to every subgame is a Nash equilibrium of that subgame.

Proposition 9.1 (SPE Refines NE)**.** *Every subgame perfect equilibrium is a Nash equilibrium.*

Proof. The entire game is itself a subgame (beginning at the root). If s^* is an SPE, then it is a NE of the entire game by definition. □

Proposition 9.2 (SPE in the Entry Game)**.** *The Entry Game (Example 8.1) has a unique SPE:* (E, A) *(Enter, Accommodate) with payoffs* $(1, 1)$.

Proof. The game has two subgames: the entire game (rooted at the empty history) and the subgame starting after player 1 chooses E (where player 2 decides between F and A).

In the subgame after E: player 2 chooses between F (payoff -1) and A (payoff 1). The unique NE of this subgame is A.

Given that player 2 plays A in the subgame, player 1 in the full game chooses between O (payoff 0) and E (payoff 1, since player 2 accommodates). Player 1 chooses E.

The unique SPE is (E, A). The non-credible equilibrium (O, F) is eliminated because F is not a NE in the subgame after entry. □

9.3 Backward Induction

Definition 9.3 (Backward Induction)**.** For a finite game with perfect information, *backward induction* is the following procedure:

1. Identify all "penultimate" nodes: non-terminal histories h such that all successors (h, a) are terminal. At each such h, the player $P(h)$ chooses the action maximizing their payoff: $a^*(h) = \operatorname{argmax}_{a \in A(h)} u_{P(h)}(h, a)$.
2. Replace each penultimate node h with a terminal node having payoff $u(h, a^*(h))$.
3. Repeat until the root is reached.

The resulting strategy profile specifies an action at every node of the original tree.

Theorem 9.3 (Backward Induction Yields SPE)**.** *In a finite game with perfect information and no ties (i.e., the payoff function is injective when restricted to any player's choices at any node), backward induction produces the unique subgame perfect equilibrium.*

Proof. We prove by strong induction on the depth d of the game tree (the maximum number of actions in any history).

Base case ($d = 1$): The game consists of a single player choosing an action at the root, with all successors being terminal. The backward induction strategy (choose the payoff-maximizing action) is trivially the unique NE and SPE.

Inductive step: Assume the result holds for all games of depth $< d$. Consider a game Γ of depth d with perfect information and no ties.

At the root, player $P(\emptyset)$ chooses action $a \in A(\emptyset)$. Each action a leads to a subgame Γ_a of depth $< d$. By the inductive hypothesis, each Γ_a has a unique SPE s_a^* with a well-defined payoff vector $v(a)$.

In the full game, backward induction assigns to the root the action $a^* = \operatorname{argmax}_{a \in A(\emptyset)} v_{P(\emptyset)}(a)$, which is unique (no ties).

We show the resulting strategy profile s^* is the unique SPE.

s^* is an SPE: In any subgame Γ_a, the restriction of s^* is s_a^*, which is an SPE of Γ_a by the inductive hypothesis. It remains to check that no player wants to deviate at the root. But the root player $P(\emptyset)$ anticipates the SPE outcome in each subgame (by the inductive hypothesis, the SPE outcome is unique), and a^* maximizes over these outcomes. The other players do not move at the root.

Uniqueness: Any SPE $\hat{s}$ must restrict to the unique SPE in each subgame Γ_a (by the inductive hypothesis), so the root player faces the same payoffs $v(a)$ and, with no ties, must choose a^*. Hence $\hat{s} = s^*$. □

9.4 Zermelo's Theorem

Zermelo's Theorem

Theorem 9.4 (Zermelo, 1913)**.** *Every finite extensive-form game with perfect information has a pure strategy subgame perfect equilibrium. If the game also has no ties (each player has distinct payoffs at all terminal nodes reachable from any decision node), the SPE is unique.*

Proof. **Existence.** We prove by strong induction on the number of non-terminal histories $|H \setminus Z|$.

Base case: If $|H \setminus Z| = 1$, the game has a single decision node (the root) followed by terminal nodes. The player at the root has at least one action maximizing their payoff (the strategy set is finite and non-empty). Any such action constitutes a pure strategy SPE.

Inductive step: Let Γ have $|H \setminus Z| = k > 1$. The root player $P(\emptyset)$ has actions $A(\emptyset) = \{a_1, \ldots, a_l\}$, and each action a_j leads to a subgame Γ_{a_j} with strictly fewer non-terminal histories. By the inductive hypothesis, each Γ_{a_j} has a pure strategy SPE $s_{a_j}^*$ with payoff vector $v(a_j) \in \mathbb{R}^n$.

Define the strategy profile s^* as follows: at the root, choose $a^* \in \operatorname{argmax}_{a_j} v_{P(\emptyset)}(a_j)$ (this set is non-empty since l is finite). In each subgame Γ_{a_j}, follow $s_{a_j}^*$.

We verify s^* is an SPE. For any subgame not containing the root, the restriction of s^* is $s_{a_j}^*$ for the appropriate j, which is an SPE by hypothesis. For the entire game (which is also a subgame): the root player is choosing optimally given the SPE continuations (by construction), and no other player moves at the root, so no player has a profitable deviation. Hence s^* is an SPE.

Uniqueness under no ties. This follows from the backward induction argument in Theorem 9.3: at each node, the argmax is unique when payoffs are distinct, so the SPE is unique. □

Corollary 9.5 (Chess Has a Determined Outcome)**.** *In chess (viewed as a finite game with perfect information — the 50-move rule and threefold repetition ensure finiteness), exactly one of the following holds: White has a winning strategy, Black has a winning strategy, or both players can force a draw.*

Proof. Chess is a two-player, finite, perfect-information game with three outcomes: White wins, Black wins, or draw. By Zermelo's theorem, a pure strategy SPE exists. In this SPE, the outcome is determined: it is either a White win, a Black win, or a draw. The SPE strategy for the favored player (or both in the draw case) is the "winning" (or "drawing") strategy that guarantees this outcome regardless of the opponent's play. □

Backward Induction and Dynamic Programming

Backward induction in game theory is the multi-agent analog of dynamic programming in single-agent MDPs. In both cases, we work backward from terminal states, computing optimal decisions at each stage given the optimal continuation. The difference is:

- **Single-agent DP**: The Bellman equation $V(s) = \max_a[r(s,a) + \gamma V(s')]$ has a unique optimal value function (contraction mapping theorem).
- **Multi-agent backward induction**: At each node, the "optimal" action depends on the other player's SPE strategy, making it a game-theoretic rather than optimization problem. With perfect information and no ties, the SPE is still unique. With imperfect information or ties, multiplicity arises.

AlphaGo and AlphaZero use MCTS (Monte Carlo Tree Search) to approximately perform backward induction in game trees too large for exact computation.

9.5 Behavioral Strategies vs. Mixed Strategies in Extensive Form

Remark 9.2 (Kuhn's Theorem Revisited). Theorem 8.2 establishes that in games with perfect recall, we can work with behavioral strategies (RL policies) without loss of generality. This is crucial for computational purposes: a behavioral strategy specifies a distribution at each information set independently, requiring $O(\sum_I |A(I)|)$ parameters, whereas a mixed strategy over pure strategies may require exponentially many parameters (one probability for each complete contingent plan).

Proposition 9.6 (Failure Without Perfect Recall)**.** *In games without perfect recall, mixed and behavioral strategies may yield different sets of reachable outcome distributions. There exist games without perfect recall in which a mixed strategy NE exists but no behavioral strategy NE does.*

Proof. Consider a one-player game where the player moves twice but forgets their first action at the second move. Specifically: the player first chooses L or R, then (in a single information set containing both post-L and post-R histories) chooses A or B. Payoffs:

- (L, A): payoff 1, (L, B): payoff 0,
- (R, A): payoff 0, (R, B): payoff 1.

A behavioral strategy specifies $p = \mathbb{P}(\text{choose } L)$ at the first node and $q = \mathbb{P}(\text{choose } A)$ at the second information set (which contains both post-L and post-R). The expected payoff is $pq + (1-p)(1-q) = 1 - p - q + 2pq$. The maximum over behavioral strategies is $1/2$ (achieved, e.g., at $p = q = 1/2$).

However, the mixed strategy that plays (L, A) with probability $1/2$ and (R, B) with probability $1/2$ achieves expected payoff $1/2 \cdot 1 + 1/2 \cdot 1 = 1 > 1/2$. This mixed strategy has no realization-equivalent behavioral strategy, since the behavioral strategy cannot correlate the first and second moves. □

10 Exercises

Exercise 10.1 (Pure NE Enumeration)**.** Find all pure strategy Nash equilibria of the following game:

	L	C	R
T	$(2,1)$	$(0,4)$	$(1,2)$
M	$(3,2)$	$(1,1)$	$(0,3)$
B	$(1,0)$	$(2,3)$	$(3,1)$

For each NE, verify the best response condition.

Exercise 10.2 (IESDS Application)**.** Apply iterated elimination of strictly dominated strategies to the game:

	L	C	R
T	$(3,1)$	$(0,0)$	$(1,3)$
M	$(1,2)$	$(2,4)$	$(3,0)$
B	$(0,3)$	$(4,2)$	$(2,1)$

At each step, identify the dominated strategy and the strategy that dominates it. State the surviving strategy profiles.

Exercise 10.3 (Mixed NE in 2×2 Games)**.** Compute all Nash equilibria (pure and mixed) of the Stag Hunt (Example 2.4). Verify the indifference principle for the mixed NE.

Exercise 10.4 (Minimax Computation)**.** For the zero-sum game with payoff matrix

$$A = \begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix},$$

compute the value of the game and the optimal mixed strategies for both players. Formulate the problem as a linear program and verify the minimax theorem directly.

Exercise 10.5 (Correlated Equilibrium Construction)**.** Consider the game of Chicken:

	D	S
D	$(-5,-5)$	$(2,-1)$
S	$(-1,2)$	$(0,0)$

(a) Find all pure and mixed Nash equilibria.
(b) Construct a correlated equilibrium that Pareto-dominates the mixed NE.
(c) Verify that your CE satisfies the incentive-compatibility constraints.

Exercise 10.6 (Backward Induction)**.** Consider the following three-stage perfect-information game: Player 1 chooses A or B. If A, player 2 chooses C or D. If B, player 2 chooses E or F. If D, player 1 chooses G or H. Terminal payoffs (player 1, player 2):

- $A \to C$: $(3,2)$, $A \to D \to G$: $(4,1)$, $A \to D \to H$: $(1,3)$,
- $B \to E$: $(2,4)$, $B \to F$: $(5,0)$.

Apply backward induction to find the SPE. List the complete strategy for each player.

Exercise 10.7 (Non-Credible Threats)**.** Consider a modified entry game: an entrant chooses E or O. If E, the incumbent chooses F, A, or P (where P means "price war"). Payoffs:

- O: $(0,3)$, (E,F): $(-2,-1)$, (E,A): $(1,1)$, (E,P): $(-3,-2)$.

(a) Find all pure strategy NE of the normal-form representation.
(b) Identify which NE involve non-credible threats.
(c) Find the unique SPE.

Exercise 10.8 (Kuhn's Theorem Application)**.** In a two-player signaling game, player 1 (the sender) observes a private type $t \in \{H, L\}$ (each equally likely) and sends a message $m \in \{h, l\}$. Player 2 (the receiver) observes the message but not the type, and chooses an action $a \in \{U, D\}$. Player 1 has four pure strategies (mapping types to messages).

(a) How many pure strategies does player 1 have? Player 2?
(b) Write down the normal-form representation.
(c) Verify that this game has perfect recall.
(d) Construct a behavioral strategy for player 1 and its realization-equivalent mixed strategy.

Exercise 10.9 (Properties of NE)**.** Prove or disprove the following statements for finite games:

(a) Every game with a dominant strategy equilibrium has a unique NE.
(b) If IESDS yields a unique strategy profile, it is a NE.
(c) The set of NE is always convex.
(d) In a two-player zero-sum game, if (x^*, y^*) and (x^{**}, y^{**}) are both NE, then (x^*, y^{**}) is also a NE.

Exercise 10.10 (SPE Existence Without Perfect Information)**.** Construct a two-player extensive-form game with imperfect information that has:

(a) Exactly one subgame (the entire game).
(b) A NE that is not "sequentially rational" at the non-singleton information set.

This illustrates why SPE has limited refinement power in games with imperfect information, motivating stronger concepts like sequential equilibrium.

11 Summary of Key Results

Result	**Statement**	**Ref.**
Order Independence of IESDS	In finite games, the set of strategies surviving iterated elimination of strictly dominated strategies is independent of the elimination order.	Thm. 2.2
NE as Mutual Best Response	s^* is a NE iff $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for all i.	Prop. 3.1
Indifference Principle	In a mixed NE, all pure strategies in the support yield equal expected payoff; strategies outside the support yield weakly less.	Thm. 4.1
Nash's Existence Theorem	Every finite game has at least one mixed strategy NE. Proved via Kakutani's fixed point theorem.	Thm. 5.2
Kakutani's Fixed Point Thm	A uhc correspondence with non-empty convex values on a compact convex set has a fixed point. Proved via Brouwer.	Thm. 5.1
Von Neumann's Minimax Thm	$\max_x \min_y x^\top Ay = \min_y \max_x x^\top Ay$ for any matrix A. Proved via LP duality.	Thm. 6.2
NE in Zero-Sum Games	NE strategies are interchangeable and all yield the game value.	Cor. 6.3
NE $\subseteq$ CE	Every NE (as a product distribution) is a correlated equilibrium.	Thm. 7.1
CE via LP	The CE set is a convex polytope; optimizing over it is an LP.	Prop. 7.2
Kuhn's Theorem	With perfect recall, mixed and behavioral strategies are realization-equivalent.	Thm. 8.2
SPE Refines NE	Every subgame perfect equilibrium is a Nash equilibrium.	Prop. 9.1
Backward Induction $\Rightarrow$ SPE	In finite perfect-information games with no ties, backward induction yields the unique SPE.	Thm. 9.3
Zermelo's Theorem	Every finite perfect-information game has a pure strategy SPE. Unique if no ties.	Thm. 9.4

Looking ahead. This lesson establishes the static and single-shot extensive-form game theory needed for multi-agent RL. The next game theory topics — repeated games, stochastic games, and mechanism design — build directly on these foundations. In Phase 02 (Core RL), the MDP framework can be viewed as a single-agent "game against nature." In Phase 05 (MARL), stochastic games combine MDPs with the multi-player game-theoretic structure developed here: at each state, agents play a stage game whose solution concept (Nash, minimax, correlated) determines the Bellman operator for the multi-agent value function.